



Drawings of the Self During COVID-19 Reveal Lived Experiences of the Pandemic

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Abstract

The early days of the COVID-19 pandemic provided a unique context in which to explore individuals' understandings of the self. Absent pharmaceutical interventions, many government and public health officials enacted behavioural interventions at this time to help curb infection. In places with social distancing or lockdown policies and recommendations, many aspects of individuals' social identities were altered, taken away, or limited. The present study uses a qualitative drawing methodology to thematically analyze individuals' perceptions of the self. Between July and December 2020, 78 participants submitted a visual in response to the prompt, "Draw how you see yourself during the COVID-19 pandemic", with the option to add a brief written description. Analysis of the images and their accompanying descriptions resulted in four themes that highlight both positive and negative self-perceptions in the areas of personal well-being, a shared sense of isolation, pandemic-related protocols, and perceptions of time. This study offers insight into the diverse ways individuals understood themselves in the early stages of the COVID-19 pandemic and highlights the value of qualitative, visual methodologies for elucidating less tangible processes and experiences.

Keywords COVID-19 · Well-being · Self · Qualitative research · Visual methodologies

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The societal and psychological impacts of COVID-19 have been broadly documented from a variety of angles, including assessments of both economic (e.g. Lenzen et al. 2020) and individual (e.g. Findlay et al. 2020) well-being. Different pandemic experiences and challenges were faced by specific populations, such as workers (Nieuwenhuis and Yerks 2021), parents (Fontanesi et al. 2020), university students (Benson and Whitson 2021), and older adults (Fuller and Huseth-Zosel 2021; Ishikawa 2020). Although there has been some theoretical work on the effects of COVID-19 on the self (e.g. Scalabrini et al. 2021), there remains significant opportunity to explore the lived experiences of the self within this context. The present study offers a qualitative analysis of conceptions of the self within the early contexts of COVID-19 based on drawings and accompanying written descriptions collected between July and December 2020, including implications for personal well-being, a shared sense of isolation, pandemic-related protocols, and perceptions of time. This timeframe, falling shortly after the classification of COVID-19 as a pandemic by the World Health Organization (WHO) on March 11, 2020, presents a significant moment for considering self-perceptions in the context of heightened uncertainty and change (Scholz and Freund 2021).

Positive and negative effects of the pandemic

Although definitions of the stages of the COVID-19 pandemic continue to evolve, terminology like “first wave” (Naughton et al. 2021) or “early” (Scholz and Freund 2021) pandemic is used to describe the initial months studied here. This period is somewhat unique due to the lack of COVID-specific pharmaceutical interventions, as well as especial uncertainty about how the pandemic would unfold (Gotlib 2021). As such, government and public health officials worldwide largely depended on behavioural interventions to attempt to control infection rates (Scholz and Freund 2021). Interventions ranged from public communication and distancing guidelines to stricter lockdowns (Jørgensen et al. 2021). Globally, interventions included actions such as stay-at-home mandates, quarantine, closure of schools, restaurants, and stores; restrictions on non-essential travel and socialization; and the adoption of mandatory personal protective equipment (Haug et al. 2020).

Although these measures were taken to reduce COVID-19 transmission, isolation negatively influenced well-being. For example, previous research suggests that isolated individuals reported decreases in physical activity levels (Chtourou et al. 2020), sleep disruptions (Salehinejad et al. 2020), eating disruptions (Galali 2020), and increased rates of smoking, alcohol intake, screen time, and distress (Lange and Nakamura 2020). Some reports have noted an increase in mental health concerns, with one UK study suggesting that the prevalence of anxiety and depression both increased by more than 25% following the pandemic’s onset (Dettmann et al. 2022). Precipitating factors include social media exposure (Gao et al. 2020) and loneliness (Killgore et al. 2020), with rates of observed loneliness increasing significantly throughout the early pandemic (Fuller and Huseth-Zosel 2021; Luchetti et al. 2020). However, experiences varied across geographical and social locations (Dettmann et al. 2022). Vulnerable populations such as immunocompromised individuals demonstrated greater feelings of stress and anxiety (Salameh et al. 2020). Similarly, older adults’ quality of life decreased as successive medical conditions went untreated (Ishikawa 2020). The mental health care needs of 18 to 25-year-olds also increased during lockdown measures (Marchini et al. 2020).

Of course, the conditions and outcomes experienced because of the early pandemic were not exclusively negative. For example, Klaver and Lambrechts (2021) suggest that requirements to work from home resulted in an increased sense of autonomy and a reduced pressure to be productive, promoting the pursuit of creativity, self-development, and personal goals. Individuals have also reported increased opportunities to strengthen familial relationships (Evans et al. 2020), connect with nature (Pouso et al. 2021; Venter et al. 2020), and engage with leisure activities (Klaver and Lambrechts 2021). Although the extent to which individuals reported positive experiences appears to vary, the existence highlights the diversity of early pandemic experiences.

The self in the pandemic

The self can be defined as the system of psychological processes that encompasses all parts of a person, including the interaction of these mechanisms with each other and the social context. The concept of the self is deeply rooted in societal and cultural contexts (Scalabrini et al. 2020), played out across the different settings and roles that people engage in throughout a lifetime (Goffman 1959). Thagard (2014) depicts the self as a multilevel interacting system with four subsystems: the social, individual, neural, and molecular levels. Disruption to the settings and roles of daily life may disrupt this self. The setting and role disruptions of the early pandemic may have had profound influences on both the social and individual levels of the self. Such influences may be best captured by assessing people's self-perceptions. Lockdown and other stay-at-home mandates presented a unique challenge to the self, stripping away and restricting both movement and social connection, resulting in limited in-person contact and interactions mediated by technology such as news reports, Zoom calls, and social media. The pandemic also made social gatherings feel like a potential threat to both self and others (Massaccesi et al. 2021). Larger existential themes, such as fear, (dis)continuity, authenticity, and freedom, are prominent features in work surrounding the self in the context of COVID-19. For example, Scalabrini et al. (2021) highlighted the ways in which Western cultural values of freedom and independence were undermined by the pandemic, noting the tension between clashes over government mandates and the sense of responsibility to self and others in managing the risk of infection. The disruption of salient social roles—friend, parent, romantic partner, employee—also influenced changes to one's sense of self during the early pandemic (Liu et al. 2021). Changes to valued roles contributed to people feeling a sense of inauthenticity or incongruence, particularly when they maintained a past-focused lens emphasizing what had changed, although adapting a present-oriented approach and inhabiting new versions of these valued roles appears to mitigate some of these negative effects (Liu et al. 2021).

Drawing as a health research method

In the context of existing observations surrounding how COVID-19 impacted lives across the globe, we sought to examine views of the self throughout the early pandemic period using a drawing-based methodology grounded in phenomenology. Understanding the self in this context generates knowledge about how individuals engage with various life domains. Prior research has demonstrated that participant-generated images offer insight on participant experiences, providing access to metaphorical and non-literal understandings

that may be difficult to express through word-based data collection methods (Broadbent et al., 2006, 2009; Cross et al., 2006; Phillips et al., 2015; Reynolds et al., 2006).

Following Edgar's (1999) call to integrate artwork into health and social science research, and drawing on the long history of art and creative self-expression in their field of occupational therapy, Henare et al. (2003) used participant-generated art to examine the experience of chronic pain. Drawing-based methods have been used to investigate experiences of other health conditions as well, including headaches (Broadbent et al. 2009), heart failure (Reynolds et al. 2006), and cancer (Harrow et al. 2008). For example, Broadbent et al. (2006) asked participants to create drawings of their heart following acute myocardial infarction, with follow-ups at 3- and 6-months post-treatment. Drawings offered insight into pre-occupations and anxiety about their experiences, with drawings of increasingly large hearts over time demonstrating associations with elevated heart-focused anxiety, health complaints, and health care use (Broadbent et al. 2006). Drawings can provide important insight into lived experiences of illness identity, emotional context, and potential pre-occupations and perceptions relevant to participant experiences (Broadbent et al., 2019; Kearney & Hyle, 2004). Further, drawings provide an accessible mode of expression by being an easy and familiar task to most adults, avoiding the technical limitations of other visual methods (Woodgate et al., 2014) regardless of potential hesitations about artistic abilities (Guillemin & Drew, 2010; Kearney & Hyle, 2004).

Drawings can also demonstrate key insights into *self*-perceptions among participants. For example, participants in Thorpe et al.'s (2019) investigation of the lived experiences of polycystic ovary syndrome used their drawings to highlight the ways in which they felt limited and controlled by their condition, noting the weight of their experiences and the ways their lives were shaped by them. Similarly, participants in Henare et al.'s (2003) study described the ways in which chronic pain contributed to feelings of loss of self and reinterpretations of the self. The incorporation of non-linguistic sources of data and the reflection required to generate them provides a rich and vibrant source of understanding such conditions (Guillemin, 2004; Guillemin and Drew 2010). Such studies have convergent validity with more traditional methods of investigation in terms of the themes and impacts highlighted by participants, thus increasing their credibility (Thorpe et al., 2019). Importantly, this open-ended visual methodology allows for the expression of mixed and ambiguous experiences, which is important for capturing effects of the pandemic, given it had such broad impacts. Through using this approach, our analysis is able to reflect areas where the pandemic impacted participants' lives, while simultaneously capturing the heterogeneity of experiences. As such, drawing-based methodology has strong potential for assessing the experience of the self during the pandemic.

Method

Participants

The study was publicly advertised on social media platforms (Reddit, Instagram, Facebook, Twitter) and to psychology research pool participants at a small university in the Pacific Northwest region of Canada. Participation was restricted to those of legal age in their place of residence. Data collection throughout the summer and fall of 2020 enabled us to adequately sample experience throughout the progression across several of the early months of the COVID-19 pandemic.

In total, 78 participants (66.7% women, 26.9% men, 5.1% nonbinary or transmasculine, 1.3% no gender indicated) returned a drawing. Average age of the participants was 24.13 (range 18–71), with 43.6% reporting being single, 24.4% in a committed relationship, 16.7% married, and 15.3% in another type of relationship. Most (64.1%) resided in Canada, while a further 19.2% reported residing in the USA; the remaining 16.7% of participants resided in 13 other countries across 5 continents (including, for example, Australia, Bermuda, and Uganda).

Procedure

In addition to demographics questions, participants were asked to reflect on and then respond to an open-ended prompt: “We would like you to draw how you see yourself during the COVID-19 pandemic”. In line with previous drawing task designs (Broadbent et al. 2004, 2006, 2009; Thorpe et al. 2019), instructions to participants were as follows:

In this project, we are interested in learning how you see yourself during the COVID-19 pandemic. Because there are many ways to experience the pandemic, people very likely also picture themselves within it in different ways. We would like you to draw how you see yourself during the COVID-19 pandemic. We are not interested in your drawing ability—a simple sketch or line drawing is fine. On a blank piece of paper, please draw a representation of your self within the pandemic. You are welcome to use any materials available to you (e.g. pencils, pens, coloured pencils, crayons, markers, etc).

Participants could also add a brief written description to help the researchers understand the meaning of the drawing from the participant’s own perspective. All images were hand-drawn on paper or on a digital device; images drawn on paper were then photographed or scanned and returned to the research team via email. Both the instructions and the procedure for returning the image were kept as simple and as open as possible to avoid limiting participant decisions about their self-representations and to help them feel confident about their ability to complete the task successfully (Guillemin and Drew 2010). Pseudonyms are used throughout.

Analytic approach

Thematic analysis of the images and the accompanying written descriptions was conducted using Rose’s (2001, 2007) critical visual methodology framework, informed by Guillemin (2004). This approach considers three meaning-making sites: the image’s production (participant-centred factors), the image itself, and the viewing of it (researcher-centred factors), and proposes several questions to guide analysis through them (Guillemin, 2004).

The lead researcher and two research assistants analyzed the images and descriptions independently and in pairs. Each image was analyzed by at least two team members, with particular attention paid to the contents of each image (both obvious and more obscure), size and positioning of the image and its elements on the page, drawing medium and use of colour, as well as connections to any written descriptions provided by the participant. Ratings were compared for agreement, with any discrepancies discussed in a team meeting. In the second step, the raters identified the central idea or ideas of each image and connected each idea to a category (for example, “feelings of anxiety” or “time urgency”). The lead

author then collapsed the categories into overarching themes. Finally, the research team met to discuss the themes and to check for consensus.

Results

Image analysis resulted in 18 categories, which were grouped into 4 descriptive themes: personal well-being, a shared sense of isolation, pandemic-related protocols, and perceptions of time. Each theme was experienced by participants differently, most often as either a benefit (a gain) or a loss, or alternately as an amplification of either relaxation or anxiety (Table 1). Drawings and descriptions were largely consistent across participant age, gender, and nationality. Younger participants were more likely to report thoughts about identity development and change and to report feelings of missing out. Male-identifying participants were somewhat more likely to reference valued activities.

Drawings most commonly portrayed the self at home, the reflective or longing self (for past and future), and the connected self; in the latter, these connections were at home, virtual, or both. Brighter shades of a range of colours were associated with experiences of excitement and joy of the self at home, thoughts of the longing self about life outside of the pandemic, the connected self spending time in person and virtually with important others. Darker shades of blue and gray indicated sadness, loneliness, difficulty, and the struggles of the longing self, with the use of black typically reserved for thoughts and concerns about death. Shades of red, orange, and yellow were the warning colours indicating feelings or thoughts about danger to self or others.

Theme 1: Personal identity and well-being

When asked to draw and reflect on the self in the COVID-19 pandemic, not surprisingly, many participants asked themselves how they were *really* doing. Experiences of mixed feelings and mood changes were represented by self-portraits featuring a split face. Drawings depicting malaise featured dark edges, with solitary, pensive figures seated at a table, desk, or alone on a bed. Clouds drawn overhead, sometimes containing words but in other instances expressed through dark scribbles, typically indicated negative emotions, particularly anxiety, stress, and feelings of being overwhelmed. Emma, age 18 (Canada; Image 1) placed a scribbled cloud close above a portrait featuring her full body wrapped around and around with cords: headphones winding down to a tablet playing a university class lecture, a cell phone with lines of writing and its charging cord wound around both arm and leg. Emma shared that the image “represents how I feel during the pandemic because I feel quite anxious about the future and now that I am in school again, I feel very overwhelmed with the amount of work that I do online...” To the right, a disembodied arm holds a clock trapped inside a cage, representing long hours that she now feels “belong to someone else”. Emma drew herself looking slightly off centre, perhaps focused on nothing physically present but instead bound to the lecture and the writing on the phone.

Change and well-being

Underpinning many aspects of the theme of personal well-being was a strong sense of change and its effects on well-being. In line with understandings that the self includes interactions with social context, participants both drew and wrote about changes in work

Table 1 Descriptive themes and examples of features

Theme	Examples of content	Examples of drawing features
Personal identity and well-being		
a. Emotional experiences	Anxiety Stress Sadness Loneliness Normal stressors amplified Malaise Feeling a toll on well-being Trying to focus on positives Toll on mental health	Solitary figures Darker shades of blue and gray Clouds and dark scribbles
b. The challenge of staying well		Split face drawings indicating mixed feelings Food items Drinking alcohol Nature—trees and birds
c. Opportunities for self-expression	Discussing silver linings Can be REAL self in pandemic Trying new hobbies Lifestyle Appearance—less attention, weight gain, hair growth Multiple demands Background distractions	Bodies with relaxed posture Images of new hobbies (e.g. juggling) Growing facial hair Casual outfits Unruly hair Students surrounded by and bound to objects
d. Changing self, changing world		
A shared sense of isolation		
a. The self at home	Feeling trapped at home Home as a refuge from chaos Feeling happy at home	Barred windows/boxes Smiling Reading Pets Bright windows
b. The reflective or longing self	Sitting and thinking Quiet Thoughts of longing for life outside the pandemic	Drawings with dark edges Alone on bed Brighter shades of colour Red, orange, and yellow indicating danger

Table 1 (continued)

Theme	Examples of content	Examples of drawing features
c. The connected self	Stuck at home but connected Small pockets of connection Spending time with others-online and in-person	Person surrounded by or within bubbles People on video calls Headphones Red hearts Bidirectional arrows
d. Missing connections	Missing social connection Lost connections Reduced social support	Prison bars Circles and slashes Xs Black and red marks
e. Technology as a driver of connection	Key driver of social connection Technology is way to talk to friends Technology is more valuable Concerns about increased screen time Technology as monotonous	Drawings of people at computers People tangled in technology Boxes
Pandemic protocols		
a. Awareness of protocols and personal protective equipment	Following guidelines Social distancing Personal engagement with protocols and personal protective equipment	Viral structures and spores Surrounded by protocols and PPE Lines and barriers Thought bubble with vaccines Masks on faces Self portraits
b. Masks as new public feature of the self	Mask as a public feature of the self Masks as a neutral characteristic	Written protocol requirements Question marks—bold and black
c. Frustration and confusion about protocols	Difficult Overwhelming Becoming tired of protocols Uncertainty about changing protocols	
d. Following rules for the sake of others	Often reported frustration or happiness with the rules Following protocols regardless of feelings for the sake of others	Maps 6-foot measurements Smiling from a distance Smoking COVID molecules

Table 1 (continued)

Theme	Examples of content	Examples of drawing features
e. Questioning the protocols	Questioning whether they were doing enough Uncertainty about what to think about the protocols Skepticism about the effectiveness of protocols	Use of question marks Angry crowd Demon
Perceptions of time		
a. Time experienced as either compassionate <i>or</i> ruthless	Having time to slow down Disorienting Overwhelming Distressing	Clocks and calendars Hypnosis spirals Use of grayscale
b. Time for rejuvenation	Mentioning family/friends Prayer Self-reflection and growth Time to sit with own thoughts	Meditating Nature Fire
c. Time to invest in valued activities	Describing hobbies Learning new things Leisure activities	Sports Video games Juggling Reading
d. Activity restriction, boredom	Need to kill time Boredom amplifies mundane daily life and routines Need to find ways to occupy self	Reclined posture Watching television, movies Swirling black lines



Image 1 (Emma, age 18, Canada)

(job loss, remote work), social life (changes to how and how often they saw friends and family), appearance (focus on comfort, long or unruly hair when salons were closed), and, among students, changes to where and how they studied. Like Emma, those participants who referenced their studies reported confusion, anxiety, and even bewilderment at their disrupted academic lives. Students frequently drew images of themselves surrounded by the “new” markers of their student life (e.g. online lectures), as well as by numerous pressures, such as background distractions and unmerciful deadlines.

Aspects of well-being

Those participants who shared the toll of the pandemic on their well-being described feelings of sadness and depression, the frustration of being separated from friends and family, and fears about what would happen next. Elizabeth (age 40, USA) depicted herself as a solitary figure on the page, crying and holding her head. Josh (age 20, Canada) drew a large, dark cloud filled with stormy black lines atop a blue head with frown and downcast eyes, and the pressures of isolation, stress, due dates, and online classes illustrated in thick

black ink outlined in urgent red. Josh shared that the isolating aspects of the pandemic had been hard on his mental health. Especially among younger participants, feelings of loneliness were often accompanied by a keen sense of missing out on important activities and potential relationships. Many felt decidedly unsettled. Fleur (age 22, The Netherlands; Image 2) represented herself as a stick figure surrounded by the people she interacts with at work or passes on the street who represent a “potential danger”, and who, in the image, are “spreading and infecting the entire page, even if you can’t really see it”. Interposed on the image are words she dwelt on throughout the pandemic, such as death (which also appears in the image as a black force at the margins of the pictured population), numbers, and government. Fleur worked as a cleaner and used transit for many of her daily activities, adding, “I really want to avoid infecting others, but I somehow keep having to go outside”. Eating, alcohol, and time spent in nature were cited by participants as approaches that they used to cope with affective distress.

Other participants reported enhancements to their personal well-being during the pandemic. Anneke (age 18, Canada; Image 3), making a comparison to her pre-pandemic self in her drawing, stated,



Image 2 (Fleur, age 22, The Netherlands)



Image 3 Anneke (age 18, Canada)

To understand how I see myself during COVID-19, means you have to look at where I was before the pandemic. I chose fire to represent myself because I fed on the alone time and mental break from school work and came out mentally stronger. I started to exercise three times a week which also made me physically stronger. I see myself as on fire because I have warmth to share again.

Some participants emphasized that the positive emotions and other benefits to their well-being came from the opportunity to be their real selves for an extended period. For Kit (age 29, USA), the pandemic meant an opportunity to lean into self-expression, adding, “I’m trans and being home and self quarantined means that I can just be myself, let my terrible pubescent chin fuzz grow and have patchy sideburns without worrying about looking professional or having other people wonder about my age and gender.”

Theme 2: A shared sense of isolation

The theme of a *shared sense of isolation* highlights a felt sense of camaraderie during major restrictions on social gatherings. This notion of connection despite being alone, or “in a bubble” with a few others, was depicted in drawings illustrating what it meant to the participant to stay at home. Many images featured solitary individuals shown in their home environment, surrounded by their virtual connections, illustrated using other boxes, bubbles, or electronic device screens containing the outside world. Daniella’s (age 18, Canada) drawing of five stick figures on a video call illustrated this isolation. Daniella explained that it “feels like everyone is trapped in a box. There’s people around you but you can’t reach them and it’s lonely”. In contrast, Chen’s (age 42, Canada) drawing highlighted the positive social aspects of telephone and video calls; her drawing interspersed bubbles of friends and family members with groupings of small red hearts showing the positive emotional connections they continued to share.

Portrayals of aloneness

Feelings of loss of connection to others, denied access to desirable places and activities, and of being trapped were illustrated with thickly drawn black or red bars, slashes, or Xs imposed over desired locations, interactions, and activities. Vineeta (age 18, Uganda) sketched a pencil drawing of herself with a large frown and sad eyes, standing trapped behind criss-crossed prison bars. Vineeta explained that she felt “captured in my house. I cannot go outside and meet my friends. Neither can I go out with my family”. Tracey (age 38, USA) depicted the ups and downs of being at home during the pandemic with two young children who “haven’t left the house to go anywhere” in a very long time. Her drawing depicted happy moments such as hugs from her children—but also concerns about their schooling, the strain of incessant messaging about mask-wearing, and her use of varying combinations of coffee, alcohol, and ice cream to get through the day. Craig (age 56, Canada; Image 4) shared that social distancing had not been particularly challenging to him, as an introvert. Nevertheless, he keenly felt the effects of his social world becoming “small, isolated, and without theatre or regular access to live music”, which Craig showed with red circles and slashes to highlight their inaccessibility. Craig traced, in bright green, the physical boundaries of his small world, comprised of two family households, with arrows indicating strong interconnections between the homes. The blue glow of television and Internet connections to the “outside world” are visible in the window of Craig’s home. Important to the image but outside of his bright home world are two “pockets of in-person” connection, Craig’s church, where he provided music support for prerecorded services, and the home of an individual with health problems for whom Craig was providing practical support.

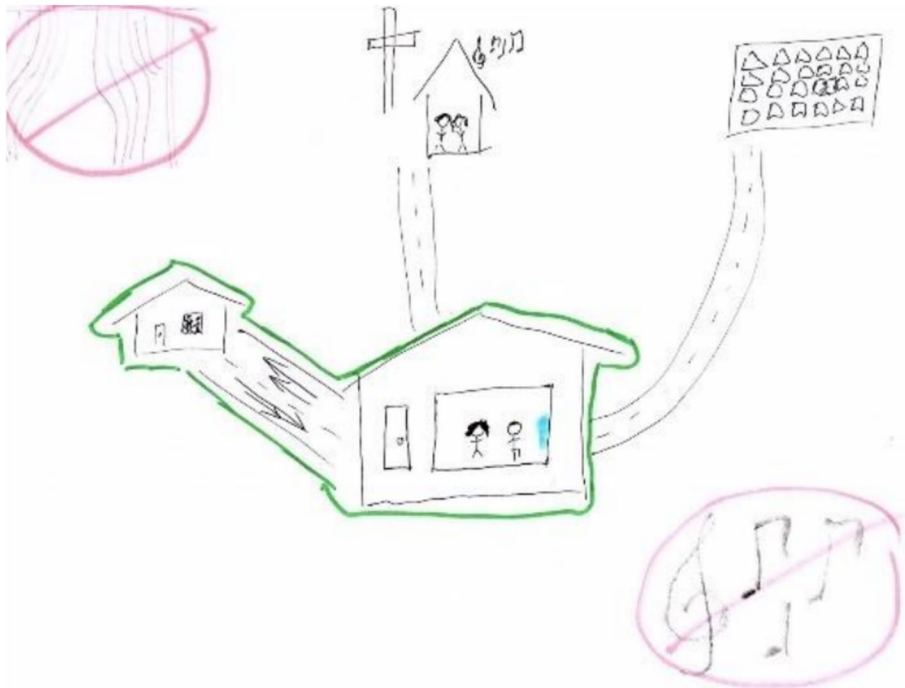


Image 4 Craig (age 56, Canada)

In contrast to those who experienced shared isolation as one of denied access or a form of imprisonment, other participants found consolation in the camaraderie of the experience and contentment in isolation. Riley (age 19; Canada) drew a bidirectional arrow between illustrations of herself and a friend. Riley shared that, although she missed spending time with those she cares about “in a normal way”, she could “still hang out with friends, but at a distance and it makes me happy knowing that I am able to protect them from the virus in some way”. Those who reported a sense of contentment in being alone drew images depicting themselves seated in a comfortable chair, often with pets, engaged in favourite activities such as reading or playing video games, and smiling. At the time of her participation, Meredith’s (age 29, USA) experience of isolation was mandated by a state-ordered 8-week quarantine. Yet Meredith experienced her home (Image 5) as a refuge from the chaos of a pandemic-altered environment, contrasting the outside, where it “looks like the world is ending”, with a scene of quiet joy inside her home. Meredith shared that the 2 months she spent off work and staying inside her home was “a time I will never forget and I will cherish. I wish I could stay in my little bubble at home forever”.

Technology and aloneness

Underpinning many aspects of the theme of a shared sense of isolation were reflections by participants on the role of technology during the pandemic. A key driver of social connection, as Emma expressed, involved being “wrapped in technology”. Maeve (age 18, Canada) noted that technology had become her only way to talk to friends. Maeve drew



Image 5 Meredith (age 29, USA)

herself looking at a computer screen during a video chat, with the self and her entire surrounding environment grayed out except for two items: a brightly coloured photo of her friends—and a set of bright pink headphones on her head (Image 6). Maeve's drawing, among others, suggested a sense that her real life had shifted to the virtual realm. Technology helped participants feel closer to others during long periods of isolation, but even as its value increased, some had concerns about the amount of time they spent looking at screens and the quality of their connections. Some participants drew themselves boxed in, limited by virtual interaction noting they felt awkward about watching others—and themselves—on a computer screen. Other drawings expressed the tedium of this technology in comparison to the stimulation experienced with face-to-face connections. Jak (age 20, Canada) emphasized the disruptiveness of the shift from his typically very active, hockey-centric lifestyle to the monotony of the daily routine of “waking up and sitting in front of the computer” depicted in his drawing.

Theme 3: Pandemic protocols

As another example of self reflecting both the psychological processes which encompass all parts of a person *and* their interactions with the social context, participants reflected on the effects of the public health protocols regarding personal protective equipment on their self-perceptions. Participants expressed an awareness of the variety of responses to public health guidelines as well as misinformation leading to skepticism about the seriousness of the virus. Li Jie (age 18, Canada) drew a demonic figure consuming him. The coronavirus



Image 6 Maeve (age 18, Canada)

zooms out of the demon, presumably to find the next victim (Image 7). He included a “house character” who repeats messages to “stay home”, a message that Li Jie shared was “felt in the back of my mind every time I went out somewhere”. Li Jie also depicted contrasting responses to public health guidelines, with voices from the crowd alternately yelling “Stay away!” and “Why tho? It’s all fake!” As a reflection on their process, Rowan (age 22, USA) stated that they were not

quite sure what emotion would be coming out on the other end, but I think it turned out a little mad-looking. I don’t get mad usually—I’m the sort of person who tends to cry rather than start a fight—but I have been furious when I look at the state of my country that has turned a horrible disease into a politicized gambit. This outbreak has matured me in a lot of ways; I wish I could say it has done the same to everybody else.

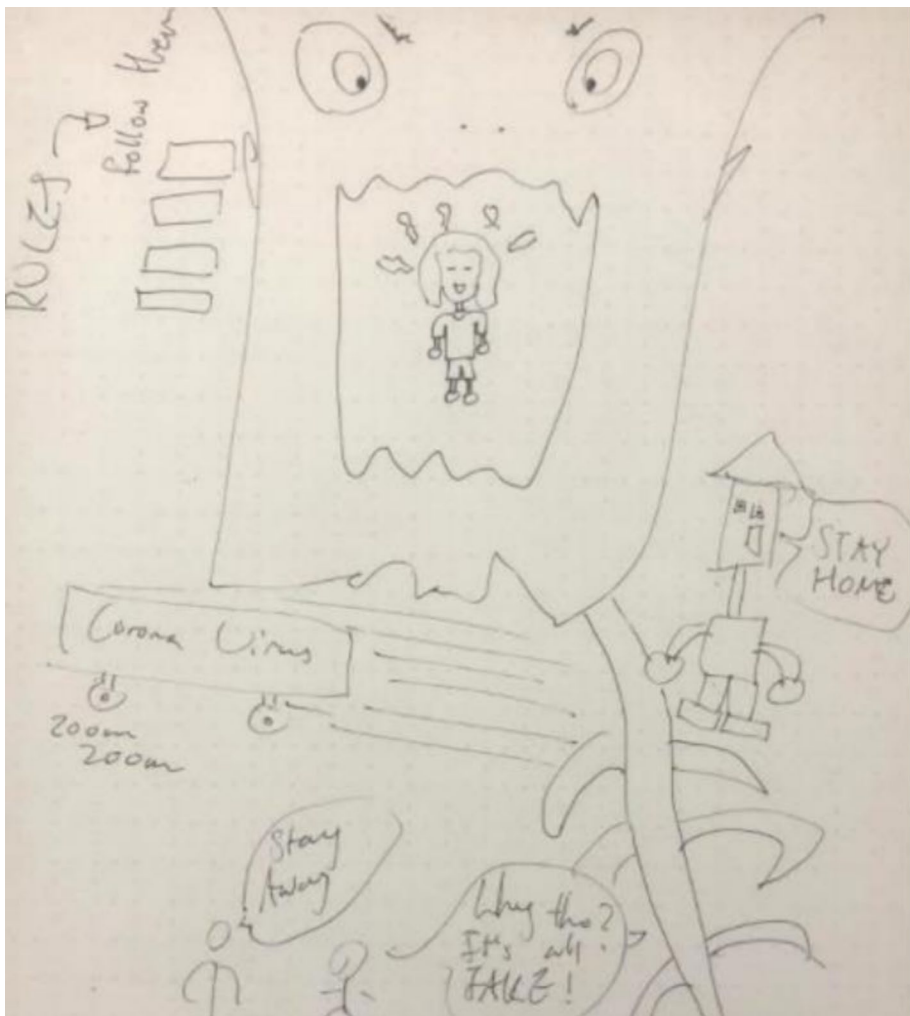


Image 7 Li Jie (age 18, Canada)

Rowan drew themselves facing straight ahead, with a large black mask tucked under thick green glasses (Image 8).

Questioning the self and the protocols

Particularly among participants in the first months of data collection, *questioning* was an underlying feature of the theme of pandemic protocols, whether the protocols were experienced as positively contributing to efforts to end the pandemic or as a frustrating experience, or both. Questioning took the form of self-questioning (“am I doing enough?”), questions about public health protocols, and skepticism about the effectiveness of those protocols. Some drawings used bold, black question marks to indicate a sense of overwhelm or of frustration with the policies that, by and large, were introduced at the outset of the pandemic and then revised over time (sometimes frequently) and by location. Others quickly tired of the pandemic and its associated protocols. Hannele (age 19, Finland) wrote, “I have become tired of the whole pandemic and a resolution seems extremely far off at this point”. Hannele drew herself alone at home, in casual attire, sitting at her kitchen table smoking a cigarette with a cloud of COVID-19 molecules, thinking about the virus, the vaccine—and vacations (Image 9).

Protocols for the good of others

Those participants who shared that they were following recommended protocols saw themselves as doing their part for the good of others, whether their personal experience with the protocols was positive, neutral, or negative. For Kiona (age 42, Canada), the difficulty with the protocols stemmed from pressure from her mother to visit. She said, “My mom is constantly mad at me for not visiting her but I’m staying away to keep her safe. She doesn’t understand and thinks I don’t love her. But it’s for her own good. I call her every day to chat for a long time.”

Following the guidelines set for their region, practicing social distancing, and wearing masks were depicted as actions that participants could undertake to minimize the chance of harm to others, particularly those at greater risk. Participant drawings also showed the ways in which the mask became a new, if temporary, public-facing feature of the self as depicted in Bethany’s (age 18, Canada) drawing (Image 10). Masks were depicted in a largely neutral manner, neither with the excitement sometimes associated with a new accessory nor with frustration or embarrassment about appearing in public wearing one. For most, mask-wearing was a marked departure from their usual public presentation but was neither a positive nor negative experience.

Theme 4: Perceptions of time

The theme of *perceptions of time* illustrates how time was alternately experienced by participants as an asset newly available for reinvestment, or as disorienting or ruthless. Participants reporting the former focused their drawings on scenes reflecting peace and contentment, sometimes with little explicit visual reference to time. Those reporting the latter made more direct references to time in their drawings. This included symbols of time such as clocks and calendars. Use of grayscale with little or no colour stood out as a common element among many of the images, illustrating confusion about time, or time standing



Image 8 Rowan (age 22, USA)



Image 9 Hannele (age 19, Finland)

still. For example, Camila's (age 24, Puerto Rico) illustration about the disorienting effects of time was drawn using a black ink pen, with no colour added (Image 11). In the image, Camila's eyes are drawn as spirals and she is surrounded by indicators of time (a calendar, a sun and moon), each accompanied by a question mark that has been drawn over with the pen multiple times, further emphasizing disorientation.



Image 10 Bethany (age 18, Canada)

Time as an asset

Drawings reflecting *time as an asset* typically featured images of nature or valued activities, often with clear connections drawn to one or more psychological processes encompassing the self. Participants described feeling rejuvenated by having more free time, or the opportunity to take a break from the “difficulty of life”. They shared about the positive emotions they derived from spending more time engaging in personal growth activities, reflecting, and at rest. Charlotte (USA) shared that “quarantine has allowed me to explore interests I haven’t really had the time to really pay attention to, like painting and birdwatching”. Charlotte drew herself outside in the long grass, surrounded by flowers and butterflies, with a bird atop her head, wearing a floral sweater reflective of her natural surroundings as if to further indicate belongingness (Image 12). Time spent in nature was understood as a key means of coping in the pandemic, as well as an activity that had real benefits for personal well-being. Dimitris (age 23, Greece)



Image 11 Camila (age 24, Puerto Rico)



Image 12 Charlotte (USA)

drew himself among tall, craggy trees. He emphasized that “Nature [has] made me feel liberated despite being closed at home away from friends. People must spend time in nature in order to find themselves”. Other participants echoed Dimitris’ comments, sharing that they were using their time to think, engage in self-reflection, pray, and realign their goals.

Like Charlotte, participants who reported experiencing *time as an asset* described using that time to invest in valued activities both new and longstanding, including leisure activities such as sports and video games. Logan (age 20, Canada) drew an image of himself practicing juggling, grinning broadly at this opportunity to “learn new things”. Logan said, “Not seeing as many people as often has definitely been a toll on my mental health, but at [the] same time I’ve had lots of time to develop myself emotionally, physically, and skill-wise, so it makes up for a lack of social stimulation”.

Burdens of time

Those who experienced time as *disorienting or ruthless* shared about distress related to activity restriction. Ongoing feelings of boredom were amplified by a sense of being restricted to mundane daily life routines. Participants reported looking for ways to “kill time” rather than deriving enjoyment or a sense of meaning from having more of it. As they sought ways to occupy themselves, they often turned to sedentary activities requiring little active engagement, such as watching television or movies. Arati (age 71, Canada) drew herself leaning back in her recliner, feet propped up, reading glasses on, and book held at just the right angle (Image 13). The rest of the page was left blank: the viewer sees only Arati, still in her chair, waiting for time to pass—just as Arati saw herself during this time.



Image 13 Arati (age 71, Canada)

Nicholas (age 25, Canada) made a pen and ink stick-figure drawing of himself sitting on a futon. There is a look of disinterest on his face as he watches a movie on his laptop computer, books and records strewn across the coffee table. He wrote, “What started out as a gift—watching TV shows I never thought I would have the time for; reading long, difficult books that demand a month of my life—now just kind of feel(s) meaningless. What’s the point of investing so much time into art if I have limited parties to discuss it with? My hobbies feel lonely now”. Nicholas concluded with this observation upon completing the drawing: “I also find it interesting that when I was asked to draw myself, I drew myself surrounded by my hobbies. Maybe with less people in my day-to-day life, I have found my identity (and) personality being of less and less meaning”.

Loss of time

The effects of perceptions of time on sense of self that Nicholas described were echoed by Morgan (age 23, Canada), who described the experience of their self in the pandemic as involving an “intense compression of time” and “thinking about all of my life, all of the time... All my thoughts on personal identity, all the changes I’ve been through, they’ve all been hitting all at once...” Morgan added, “I used to identify very strongly with my career and my religion. Both of those are coming into question and while it’s exciting it’s also terrifying. I’ve lost the primary signifiers for my personal identity”. Morgan represented their own self as a lightly pencil-drawn person with heavily drawn, swirling black lines spiralling out from their chest, surrounded by constellations (Image 14). They described their image as one of the self being “buried under this black hole that’s sucking all the constellations in” as time compresses.

Discussion

This study visualizes experiences and self-perceptions during the first months of the COVID-19 pandemic. The drawing methodology facilitated self-expression of experiential, sometimes ambiguous processes that may otherwise be difficult to articulate. The primary means of establishing validity in the present study was to encourage participants to provide an accompanying written reflection. Those reflections guided our interpretation of their images, the evidence of which lies primarily in the exemplar image but also in the participants’ own descriptions of those images (Guillemin, 2004). Because of the openness of the method, a range of experiences emerged for each study theme, with positive, negative, and ambivalent valences. Such fine-grained analysis complements previous work documenting effects of the pandemic. These divergent experiences may be mapped onto Thagard’s (2014) account of the self as a multi-level system. The drawings revealed that pandemic disruptions cascaded into both social- and individual-level self-perceptions, highlighting the interdependence of these subsystems.

The self in context

Our findings shed light on how disruption to settings and roles in the pandemic was experienced through construal of the self. Mirroring Thagard’s (2014) contention that the social and individual levels of the self are interacting aspects of the self-system, these observations highlight how the pandemic changed perceptions of both the social and individual levels.



Image 14 Morgan (age 23, Canada)

In the face of constricted access to spaces that many experienced in the early months of the pandemic (Joshi and Venkatesan, 2024), some participants wrestled with a particular tension: one can take the body out of social contexts, but one cannot take the social influence out of the self. Even in isolation, the two are intertwined. Participants clearly thought about others in the face of isolation, from concerns about infecting others to the lengths people went through to stay connected in alternate formats. Several drawings display participants'

complex and sometimes contradicting feelings around the self as a social being in a time of social isolation. The drawings reveal a complex tension: because the self is in part social, it gains much from being together with others, but also because the self is in part social, in the early pandemic social distancing could be understood to be a mechanism of mutual protection. Prior research has noted this tension between the need for social connection and the potential for infection of the self and others during the pandemic. Scalabrini et al. (2021) emphasize how existential fear manifests when there are concerns of infecting the self and others; this fear undermines the social connection that is so important for defining oneself in social contexts. Some participants' drawings and descriptions indicated fear and hesitancy to interact with others, both because of self-concerns and other-concerns. Yet, while these relational strains could have negative impacts, by and large, participants characterized their behaviours as a form of mutual aid and concern for collective well-being. Perhaps then, the lasting impact on relationships could be one of enhanced caring.

Well-being

That some participants saw their experience of the self during the pandemic as one characterized by reductions to personal well-being aligns with prior studies showing the negative effects of perceived isolation (Cacioppo and Hawkey 2009). This includes research specifically examining social isolation during the pandemic (Clair et al. 2021). However, our qualitative analysis allowed more nuanced narratives, which revealed that others derived benefits from those same experiences of isolation. The theme of *a shared sense of isolation* encapsulates both the positive and negative experiences of the self during this unique arrangement of social isolation. Participant drawings also revealed the complex nature of the relationship with self, time, and well-being in isolation. In the sudden absence of regular rhythms and routines, time could be experienced either positively as an asset (having more free time to invest as one pleased), or negatively as a source of disorientation, boredom or existential distress. These varied experiences illustrate how shifts in the social self can have differential impacts on individual-level well-being, which is consistent with perspectives that treat the self as spanning social and individual processes. While it is unclear what specifically led to such polarized interpretations of certain pandemic features, it is possible that variations in individual circumstances, such as location, support, access to information, and resource disparity, may have played a role in this regard. The diverse experiences reported by participants suggest that future research on the effects of social isolation on well-being and related constructs may benefit from the assessment of potential hidden moderators. Specifically, future research may assess whether personality traits related to social behaviour, such as extraversion, moderate the relationship between social isolation and well-being. Research might also assess whether causal attributions of one's social isolation moderate this relationship in the context of a pandemic. That is, personal attributions of fault may allow social isolation to decrease well-being, but structural attributions may mitigate against these effects. Participant reports of use of food, smoking, intoxicants, and screen time as coping strategies to navigate affective distress align with other research on isolation and home confinement, including the COVID-19 pandemic (see Lange and Nakamura 2020). However, participants also shared about spending time in nature, investing time in their hobbies, and exercising more frequently. This suggests that healthy behaviours can be adopted in place of less functional approaches. Of note, even as some used these latter activities to manage distress, in other cases, these same behaviours emerged as the result of a deliberate decision to seize what they saw as a rare opportunity, a break from ordinary life to take advantage of.

Participants grew through self-expression, strengthening their mental and physical reserves, picking up former hobbies, or learning new information and skills. Those who were afforded both the time and the means to engage in growth activities may well have been actively creating a future self that they desired before the pandemic.

Professional identities

While nearly everyone's work life was affected by changes during the pandemic, the influence on professional identity differed depending on the degree to which one's work and competence were challenged by these changes. In our sample, workers and students alike shared that some aspects of work became overwhelming. Students noted increased difficulty or anxiety about their studies because of the changes to workflow and to assignments. Confidence and familiarity with traditional structures and practices of learning were undermined by the new challenges of online learning and digital-based assignments, which seem to have challenged agentic identities for some of our student participants. Other research has highlighted how specific professional identities were challenged and underwent rapid change during the early days of the pandemic. Kröger (2020) suggested that the difficult conditions under COVID-19 undermined several key facets of health care professionals' identities, including professional expertise, resilience, and the ethical obligation to prevent death and suffering. Sahay and Dwyer's (2023) study of nursing and identity examined the unrealistic expectations encapsulated within the concept of the "healthcare hero" on the front lines. Some occupation-specific challenges were identified by our participants, as well. For example, Fleur shared about her distress that, as a cleaner, she risked infecting others. While participants in some occupational roles were able to stay isolated, and some expressed pride in doing so to keep others safe, others did not have this luxury. Future research might investigate conceptions of the self in relation to professional identities that "afford" working from home and ones that do not.

Limitations

While this investigation uses primarily nonverbal, more metaphorical means to shed light on the experience of the self in the pandemic, the study comes with some limitations. Information about socioeconomic status was not collected, which limits external validity. Analysis must be interpreted within the cultural context in which it is embedded (Henrich et al., 2010), because different cultural role structures may shape the experience of the social self with downstream influences on individual self-perception. These themes should not be taken to represent other settings with different norms and responses to the pandemic, where people may have had other experiences of lockdown and its effects on the self. The heterogeneity of experiences of the pandemic requires a broad synthesis of participant experiences, which may fail to fully represent uncommon experiences, and this problem may have an outsized effect on participants who have intersecting minority identities or circumstances, both of which are difficult to assess in our summary.

Methodological and substantive conclusions

The present study further demonstrates the feasibility of drawing methodology as an approach to qualitative research that does not require participants to be in-person or interviewed

face-to-face. This confers several advantages. It enables the methodology to be scaled to reach more participants with varied experiences without requiring huge financial or temporal resources on the part of researchers or participants. This scalability provides researchers with the opportunity to reach people outside of traditional geographical boundaries that may have different experiences than those that are within reach. In addition, collecting drawings that people created in their own homes may reinforce ecological validity, as people may more accurately represent their natural selves in their own home setting. Finally, online data collection may also remove demand characteristics that come with in-person qualitative assessments, despite potential discomfort with divulging personal information without rapport.

This study offers insights into the experience of the self during the early stages of the COVID-19 pandemic through analysis of participant-generated drawings. By exploring and contrasting the divergent ways individuals represented themselves amidst the disruptive effects of this time period, we characterized key themes related to personal well-being, shared experiences of isolation, pandemic protocols, and time. These findings highlight the potential for wide-ranging emotional and psychological impacts of a pandemic, from increased anxiety and loneliness to opportunities for personal growth and reflection. This study underscores the value of qualitative, visual methodologies in capturing complex and nuanced experiences and suggests avenues for future research on the psychological effects of such crises. The themes identify areas where the different levels of the self interact, particularly how disruptions to the social level can drive altered perceptions of the individual even in isolation. Understanding these varied self-perceptions provides new insight into the struggle and thriving that persisted in tandem across isolated individuals during the pandemic.

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Data availability Study materials and data are available from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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